PSYCH 201B

Statistical Intuitions for Social Scientists

From models to meaning

03/02/2026

Today's Plan

1. Recap
2. Marginal Effects (mixing board)
3. Final Project template time

Recap

Neyman-Pearson Approach

- Two possible states of the world
  - H0 (null) is **true**
  - H0 (null) is **false**
- Two decisions we can make
  - **reject** H0
  - **retain** H0
- Two ways we can be wrong
  - **Reject** when H0 is **true** — False Positive **Type I** error α
  - **Retain** when H0 is **false** — False Negative **Type II** error β

Imagine you're a (medical) doctor…

H_0: Not pregnant **H_1**: Pregnant

Type I Error

Type II Error

Type I Error: Falsely rejecting the null hypothesis (even though it is true)

Type II Error: Failing to reject the null hypothesis (even though it is false)

Imagine you're a (medical) doctor...

H_0: Not pregnant **H_1**: Pregnant

Ways to be wrong

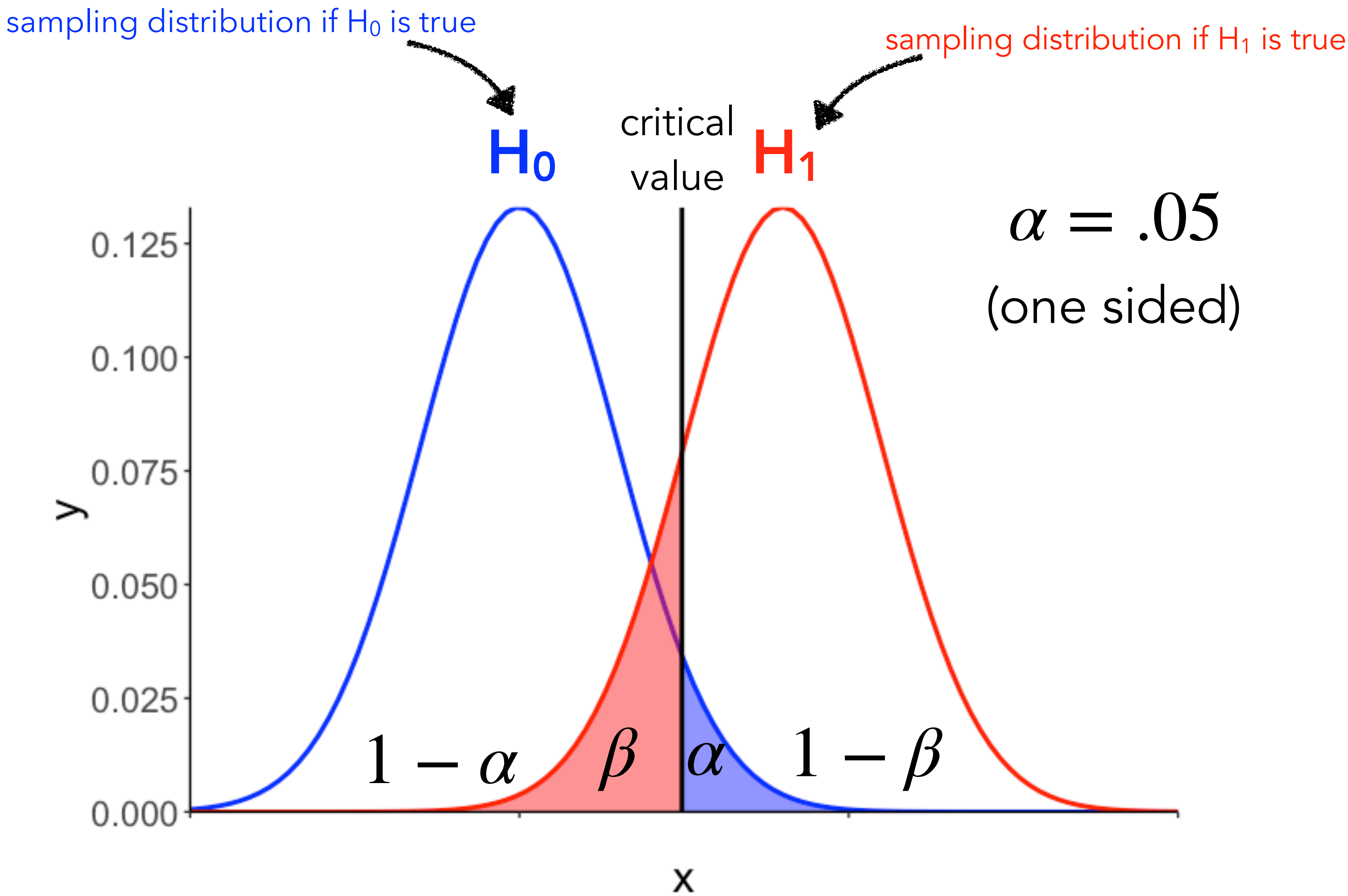
false positive	Type I error $p(\text{reject } H_0 \mid H_0 \text{ is true})$	α
false negative	Type II error $p(\text{not reject } H_0 \mid H_1 \text{ is true})$	β

Ways to be right

true positive	Sensitivity $p(\text{reject } H_0 \mid H_1 \text{ is true})$	Power $\boxed{1-\beta}$
true negative	$p(\text{not reject } H_0 \mid H_0 \text{ is true})$ Specificity	$1-\alpha$

What affects power?

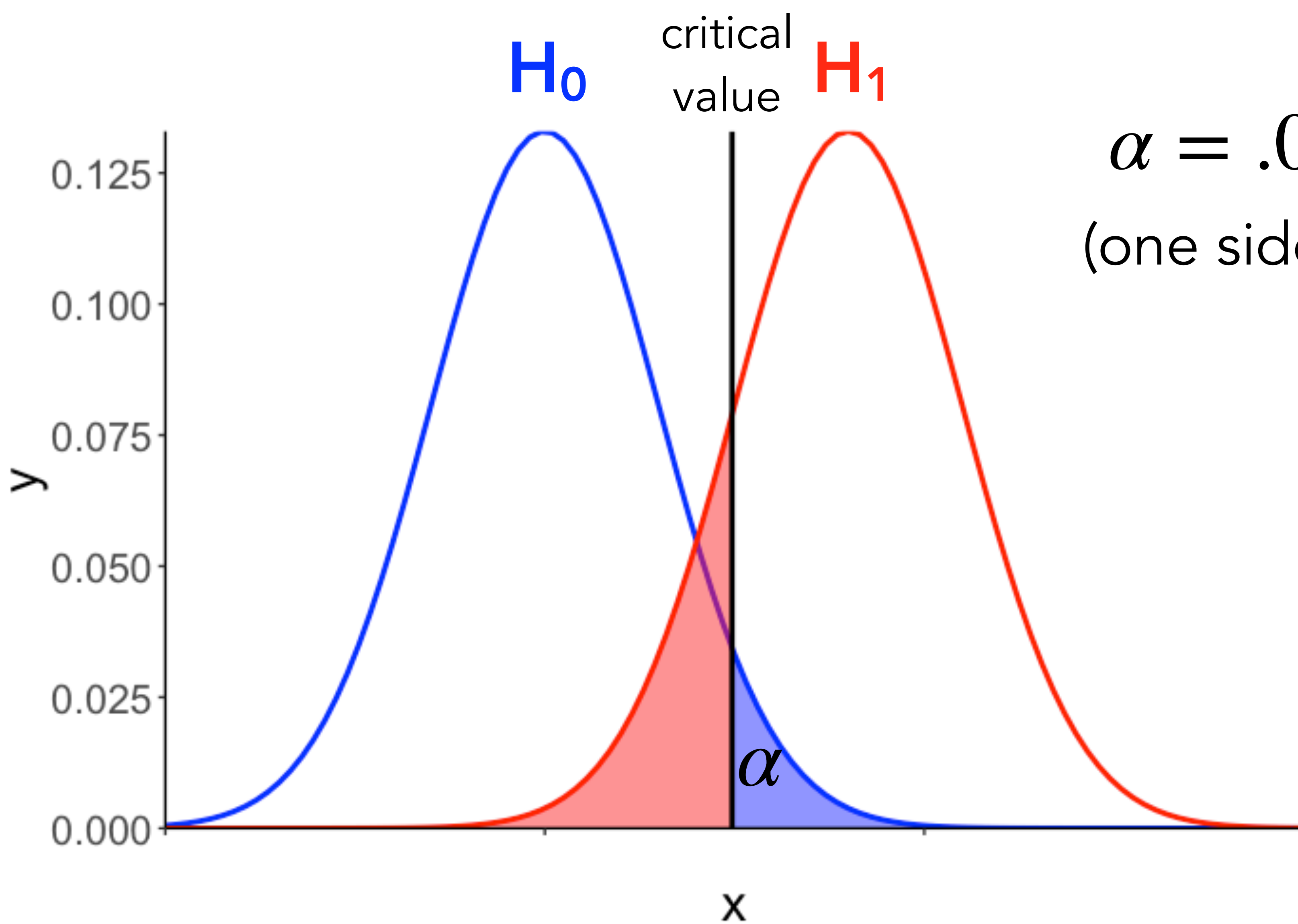
false positive	Type I error p(reject $H_0 \mid H_0$ is true)	α	true positive	Sensitivity p(reject $H_0 \mid H_1$ is true)	Power	$\boxed{1-\beta}$
false negative	Type II error p(not reject $H_0 \mid H_1$ is true)	β	true negative	p(not reject $H_0 \mid H_0$ is true) Specificity		$1-\alpha$



α affects power

(e.g. p-value threshold)

					Power
false positive	Type I error $p(\text{reject } H_0 \mid H_0 \text{ is true})$	α	true positive	Sensitivity $p(\text{reject } H_0 \mid H_1 \text{ is true})$	$\boxed{1-\beta}$
false negative	Type II error $p(\text{not reject } H_0 \mid H_1 \text{ is true})$	β	true negative	$p(\text{not reject } H_0 \mid H_0 \text{ is true})$ Specificity	$1-\alpha$

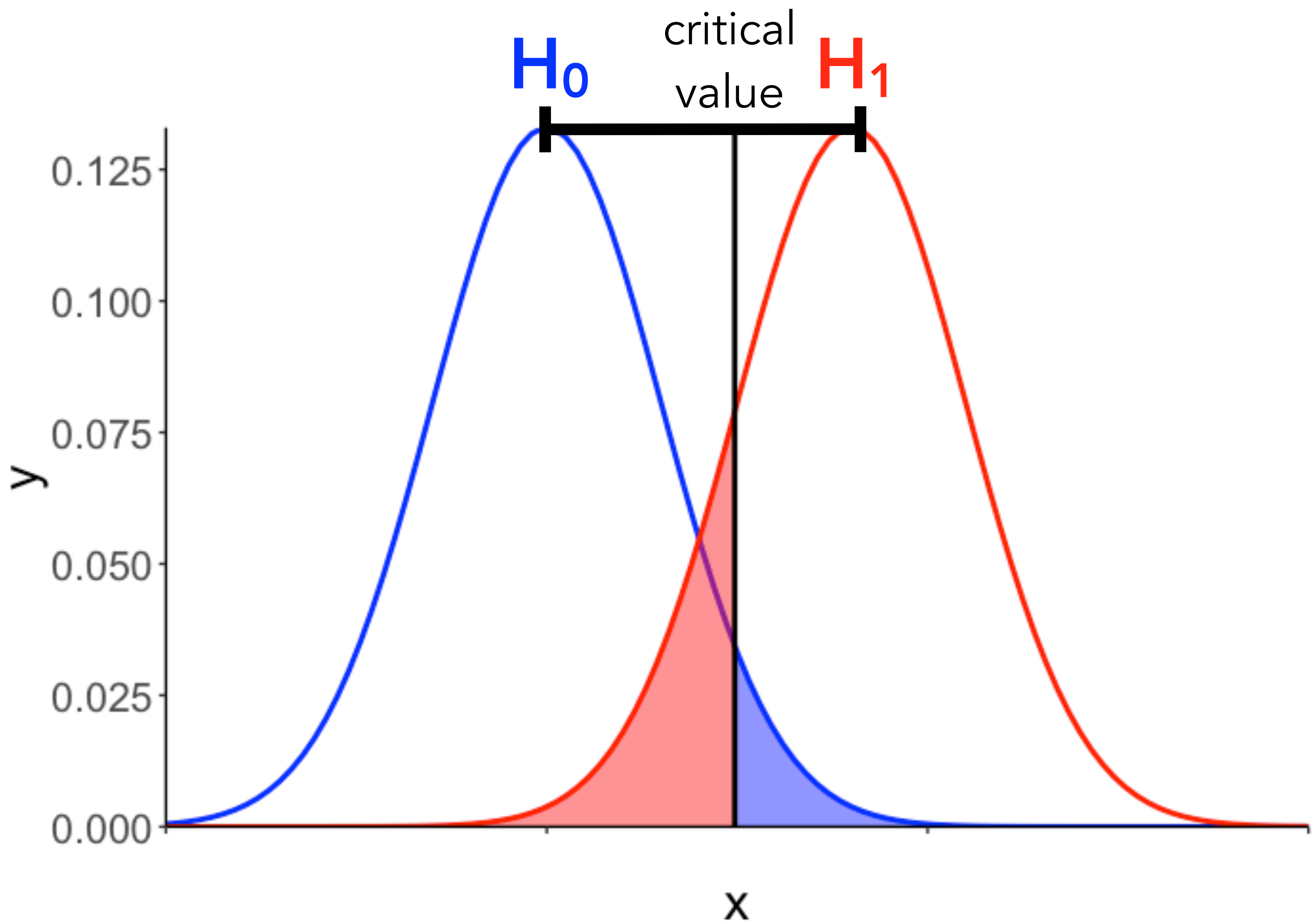


$$\alpha = .05$$

(one sided)

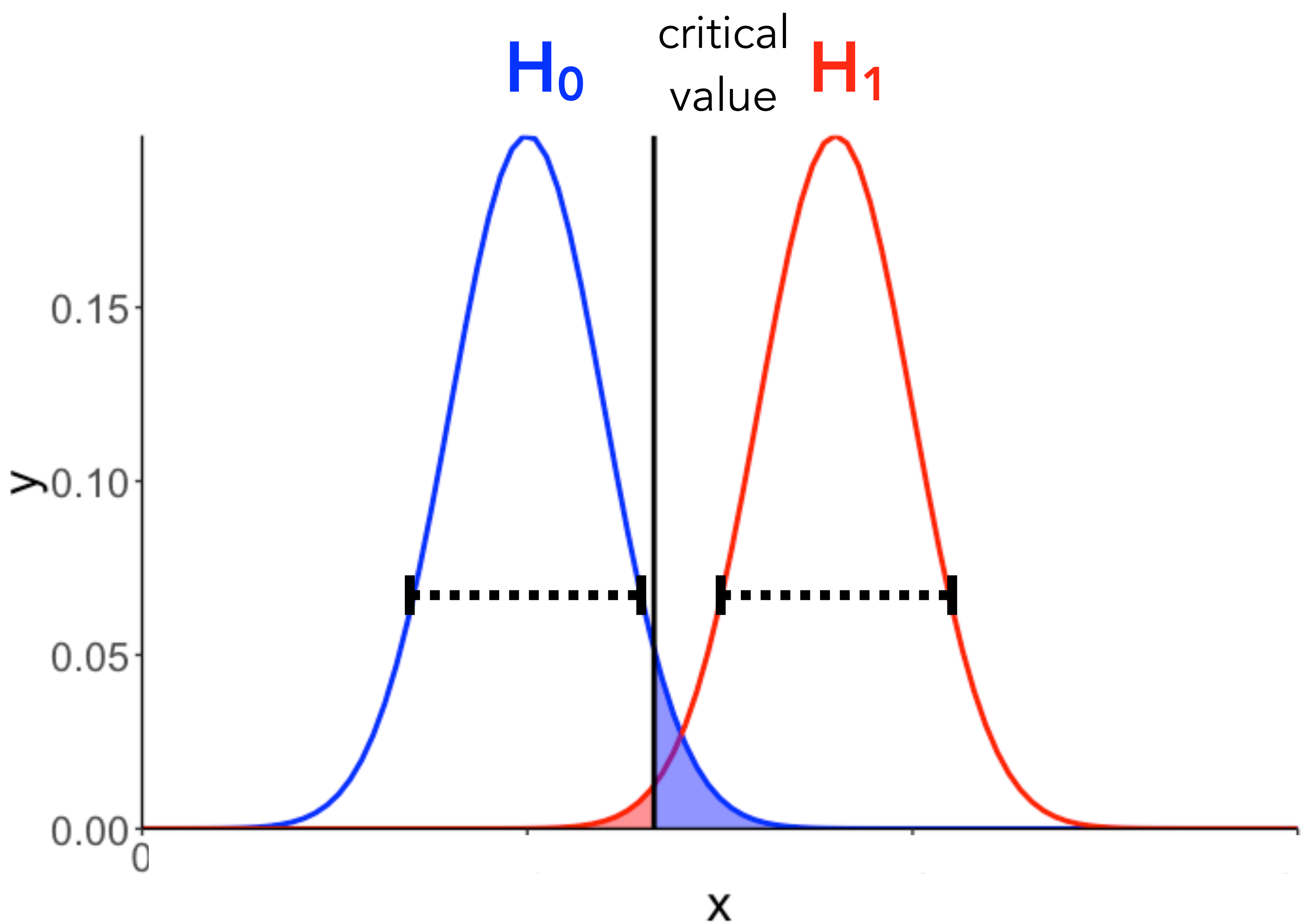
effect size affects power

false positive	Type I error $p(\text{reject } H_0 \mid H_0 \text{ is true})$	α	true positive	Sensitivity $p(\text{reject } H_0 \mid H_1 \text{ is true})$	Power $\boxed{1-\beta}$
false negative	Type II error $p(\text{not reject } H_0 \mid H_1 \text{ is true})$	β	true negative	$p(\text{not reject } H_0 \mid H_0 \text{ is true})$ Specificity	$1-\alpha$

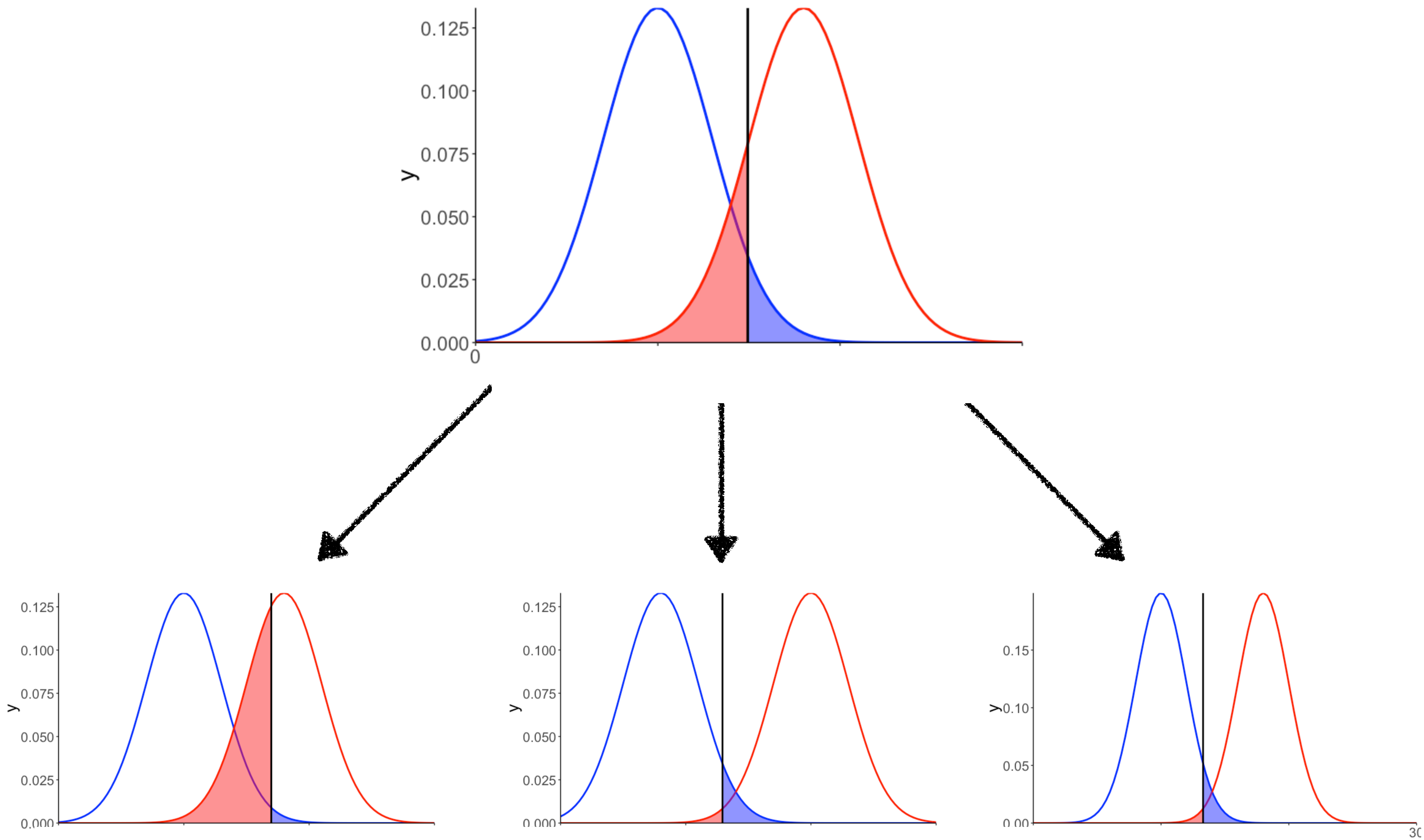


variance affects power

false positive	Type I error $p(\text{reject } H_0 \mid H_0 \text{ is true})$	α	true positive	Sensitivity $p(\text{reject } H_0 \mid H_1 \text{ is true})$	Power $\boxed{1-\beta}$
false negative	Type II error $p(\text{not reject } H_0 \mid H_1 \text{ is true})$	β	true negative	$p(\text{not reject } H_0 \mid H_0 \text{ is true})$ Specificity	$1-\alpha$



The knobs we can turn to affect power



a threshold

effect size

sample size

What α should we use? It's become conventional to set $\alpha = 0.05$. To be honest, this owes more to the fact that the resulting k tends to 1.96 as $n \to \infty$, and $1.96 \approx 2$, and most psychologists and economists could multiply by 2, even in 1950, than to any genuine principle of statistics or scientific method. A 5% error rate corresponds to messing up about one working day in every month, which you might well find high. On the other hand, there is nothing which stops you from increasing α. It's often illuminating to plot a series of confidence sets, at different values of α.

What about power? The **coverage** of a confidence set is the probability that it includes the true parameter value. This is not, however, the only virtue we want in a confidence set; if it was, we could just say "Every possible parameter is in the set", and have 100% coverage no matter what. We would also like the *wrong* values of the parameter to have a high probability of *not* being in the set. Just as the coverage is controlled by the size / false-alarm probability / type-I error rate α of the hypothesis test, the probability of excluding the wrong parameters is controlled by the power / miss probability / type-II error rate. Test with higher power exclude (correctly) more parameter values, and give smaller confidence sets.

Parameter Inference

$$t = \frac{\hat{\beta}}{SE_{\hat{\beta}}}$$

Effect / Signal

Uncertainty / Noise

ratio determines inference (p-val)

we need to account for this correctly, otherwise we're risk being too liberal with our inference aka *increasing our Type I Error (False Positive Rate)*

Parameter Inference is proportional to nested model comparison!

$$F = t^2$$

$$t = \sqrt{F}$$

```python
# Parameter t-statistic Squared
m_aug.params[1, 'statistic'] ** 2
```

0.0s

```
108.99171519122353
```

```python
# Model comparison F-ratio
results[1, 'F']
```

0.0s

```
108.9917
```

Parameter Inference Takeaways

$$t = \frac{\hat{\beta}}{SE_{\hat{\beta}}}$$

- Large t-stat or small p-value is **not** about what variables are most important for prediction!
- It's about what **signals** can be discerned from **noise**
  - *Does the addition of this parameter provide lower prediction error that what adding random noise would produce?*
- Why? Because our *uncertainty* (SE) is influence by
  - Magnitude of $(\hat{\beta})$
  - Model avg error (MSE)
  - Effective sample size (# observations vs # params aka DF)
  - Variance of each predictor $(x1 \ldots xn)$
  - Correlation between predictors (columns of X)

Remember: Statistical significance is not practical significance

Effect sizes

- a p-value tells us whether we can reject the H_0
- effect sizes is a measure of the strength of the actual effect

Why can't we just use p-values as a measure of the effect size?

$$F = \frac{\mathrm{PRE}/(\mathrm{PA} - \mathrm{PC})}{(1 - \mathrm{PRE})/(n - \mathrm{PA})}$$

any PRE will become significant if *n* gets large enough

PRE = proportional reduction in error
PA = # parameters in the augmented model
PC = # parameters in the compact model
n = sample size

Effect sizes

- a p-value tells us whether we can reject the H_0
- effect sizes is a measure of the s**trength of the actual effect**

Why can't we just use p-values as a measure of the effect size?

$$t = \frac{\hat{\beta}}{SE_{\hat{\beta}}} = \frac{\hat{\beta}}{\frac{SD_{\hat{\beta}}}{\sqrt{n}}}$$

any t will become significant if *n* gets large enough

Effect Size Reference

Guide to Effect Sizes and Confidence Intervals

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11 Variability Differences
12 Non-Parametric Effect

Guide to Effect Sizes and Confidence Intervals

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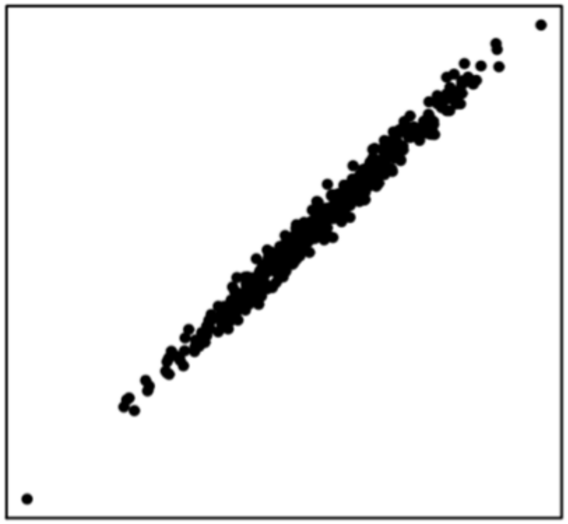
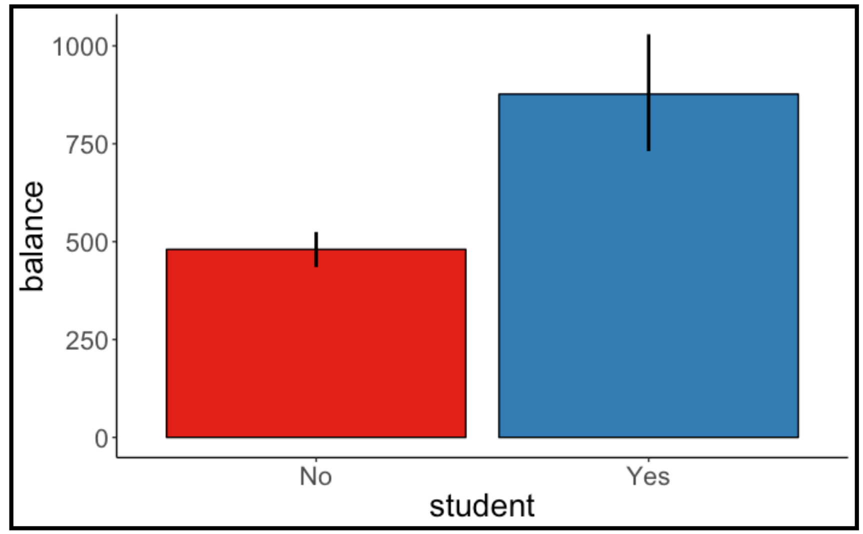
Introduction

https://matthewbjane.quarto.pub/

Common effect sizes

Relationships between variables	Differences between groups
	
r correlation	Cohen's d

Pearson correlation

$$r(X, Y) = \frac{\sum_{i=1}^{n} (X_i - \overline{X})(Y_i - \overline{Y})}{\sqrt{\sum_{i=1}^{n} (X_i - \overline{X})^2} \cdot \sqrt{\sum_{i=1}^{n} (Y_i - \overline{Y})^2}}$$

Cohen's guidelines for the social sciences

Effect size	*r*
Small	0.1
Medium	0.3
Large	0.5

depends very much on the domain

Cohen's *d*

- standardized difference between two means

absolute difference between means

$$d = \frac{|\bar{y}_1 - \bar{y}_2|}{s_p}$$

pooled standard variation

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Effect size	***d***
Very small	0.01
Small	0.20
Medium	0.50
Large	0.80
Very large	1.20
Huge	2.0

Ratio of: between variation / pooled variation

Marginal Effects

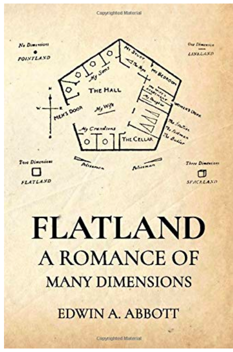
What is a model?

“A model is a logical story expressed in *math* (code)”

~ Andrew Gelman

one of the most famous statisticians;
inventor of Stan; Prof at Columbia

What is a model?



General: A general mathematical **function** that transforms *inputs* into *outputs*

$$\text{output} = f(\text{input})$$

$$\text{happiness} = f(\text{chocolate})$$

$$y = f(x)$$

$$y = f(x1,\ x2, \dots)$$

It's our theory in *math*

Jolly & Chang, 2019

What is a model?

A **theory** of how **observed data** were **generated**

```
Data = Model + Error
```

how shall we
define this?

Dataset: Credit-Card Debt

index	income	limit	rating	cards	age	education	gender	student	married	ethnicity	balance
1	14.89	3606	283	2	34	11	Male	No	Yes	Caucasian	333
2	106.03	6645	483	3	82	15	Female	Yes	Yes	Asian	903
3	104.59	7075	514	4	71	11	Male	No	No	Asian	580
4	148.92	9504	681	3	36	11	Female	No	No	Asian	964
5	55.88	4897	357	2	68	16	Male	No	Yes	Caucasian	331
6	80.18	8047	569	4	77	10	Male	No	No	Caucasian	1151
7	21.00	3388	259	2	37	12	Female	No	No	African American	203
8	71.41	7114	512	2	87	9	Male	No	No	Asian	872
9	15.12	3300	266	5	66	13	Female	No	No	Caucasian	279
10	71.06	6819	491	3	41	19	Female	Yes	Yes	African American	1350

variable	description
income	in thousand dollars
limit	credit limit
rating	credit rating
cards	number of credit cards
age	in years
education	years of education
gender	male or female
student	student or not
married	married or not
ethnicity	African American, Asian, Caucasian
balance	average credit card debt in dollars

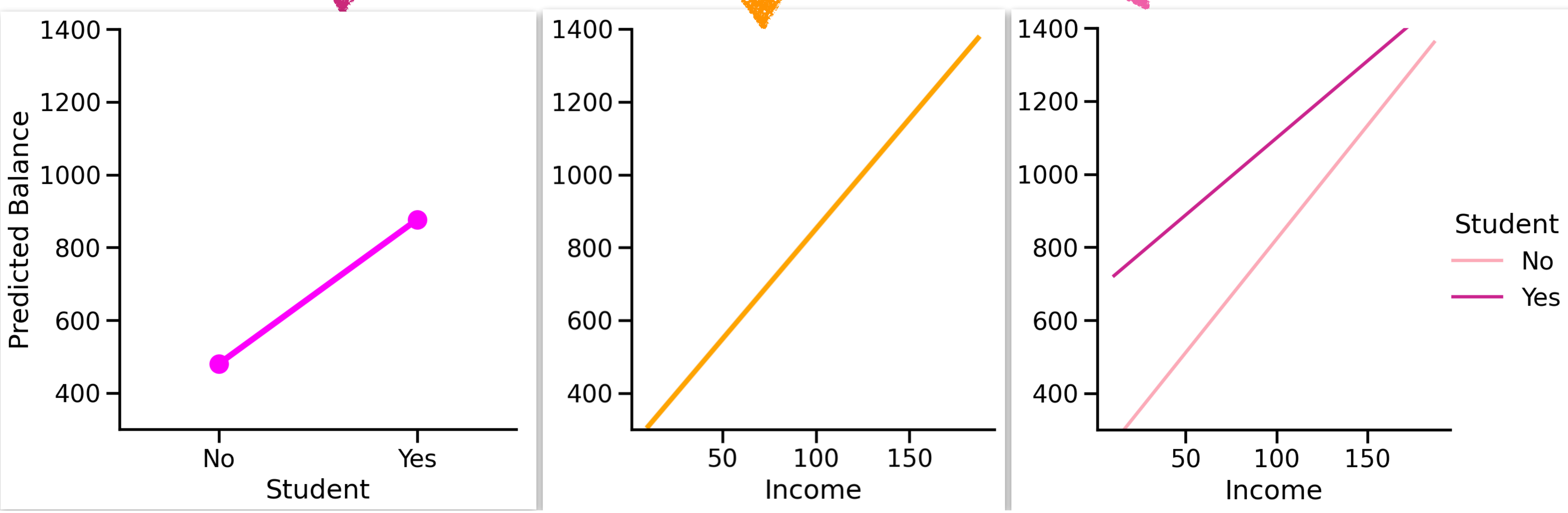
Do students have a different credit card balance from non-students when accounting for income?

Do students have a different credit card balance from non-students when accounting for income?

$$\text{balance}_i = \beta_0 + \beta_1 \text{student}_i + \beta_2 \text{income}_i + \beta_3 \text{student} * \text{income}$$

Let's unpack this visually

```
Coefficients:
                          Estimate
            Intercept      200.623
         Student[Yes]      476.676
               Income        6.218
  Student[Yes]:Income       -1.999
```



Do students have a different credit card balance from non-students when accounting for income?

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Intercept	200.623
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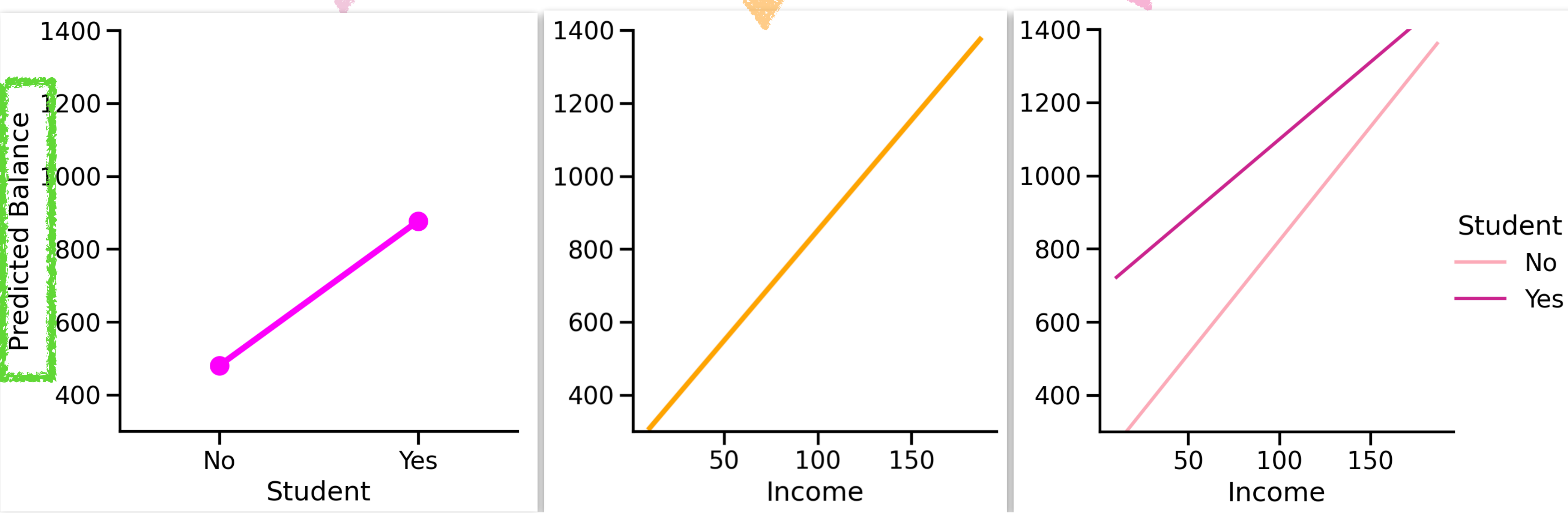
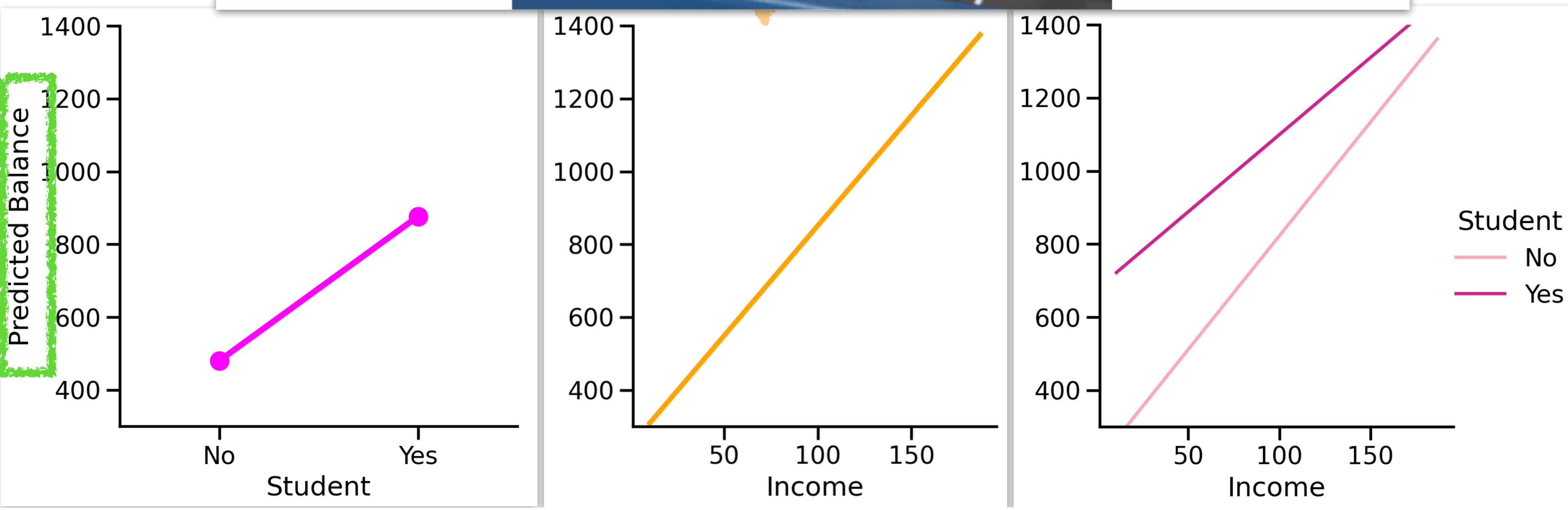


Predictions ($\hat{y}$)



Predictions ($\hat{y}$)

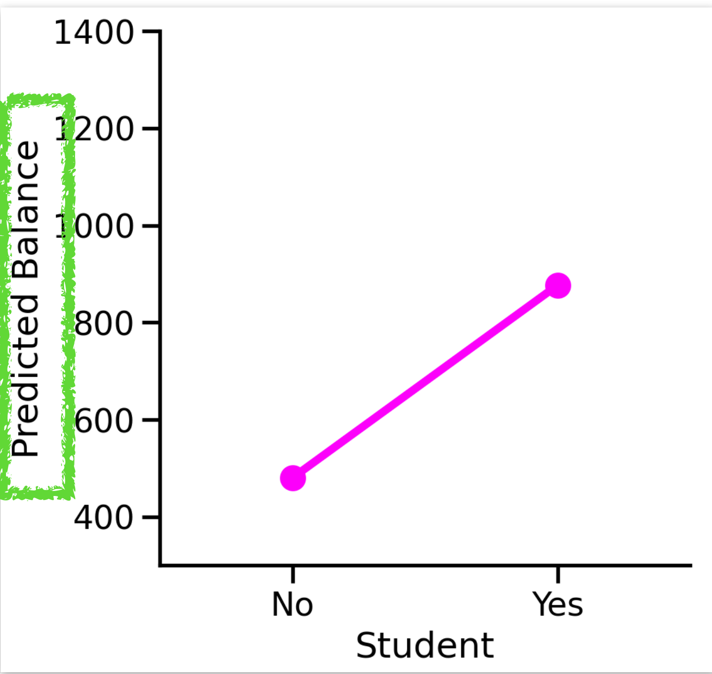
$$\text{balance}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{student}_i + \hat{\beta}_2 \text{income}_i + \hat{\beta}_3 \text{student} * \text{income}$$

We estimated (hat) these - so we can *use* them!

How does predicted balance change when we switch from Student [No] to Student[Yes]?

```
Coefficients:
                         Estimate
            Intercept     200.623
         Student[Yes]     476.676
               Income       6.218
  Student[Yes]:Income      -1.999
```

Marginal effect of student



Predictions ($\hat{y}$)

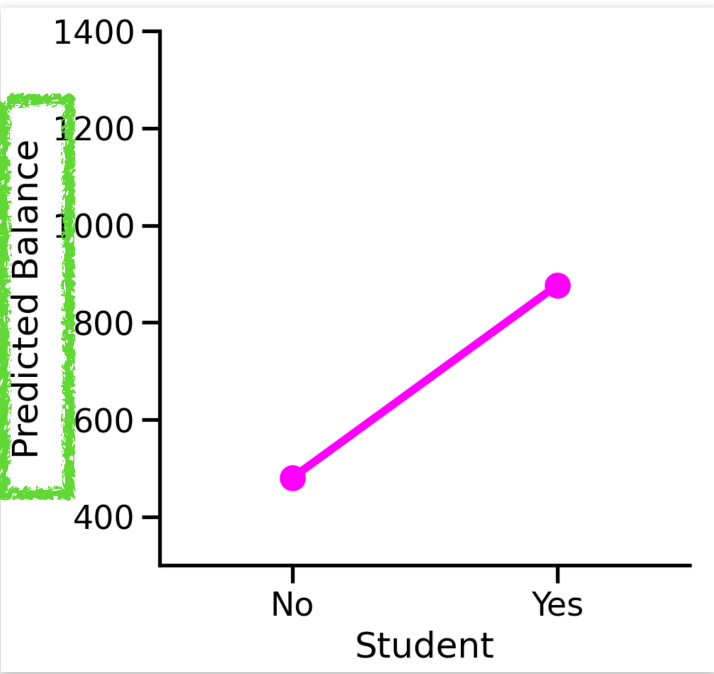
$$\text{balance}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{student}_i + \hat{\beta}_2 \text{income}_i + \hat{\beta}_3 \text{student} * \text{income}$$

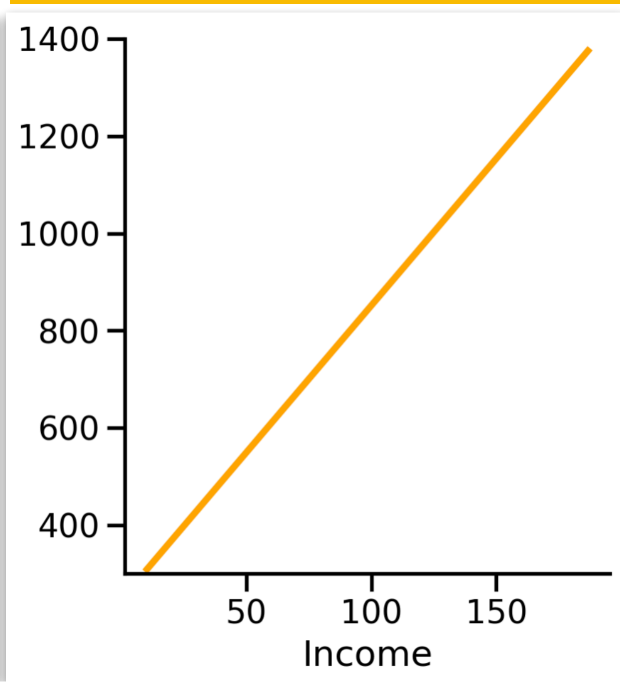
We estimated (hat) these - so we can *use* them!

How does predicted balance change when we slide Income from min to max?

```
Coefficients:
                          Estimate
            Intercept      200.623
         Student[Yes]      476.676
               Income        6.218
  Student[Yes]:Income       -1.999
```

Marginal effect of income





Predictions ($\hat{y}$)

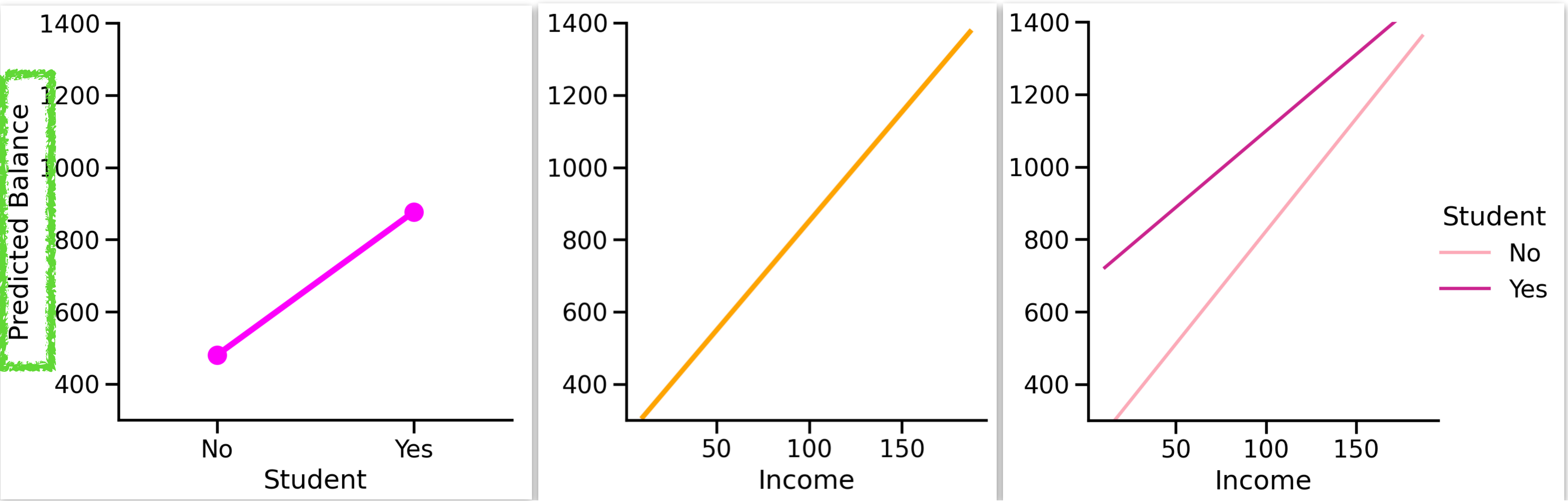
$$\text{balance}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{student}_i + \hat{\beta}_2 \text{income}_i + \hat{\beta}_3 \text{student} * \text{income}$$

We estimated (hat) these - so we can *use* them!

How does predicted balance change when we slide Income from min to max AND switch from Student [No] to Student[Yes]?

```
Coefficients:
                        Estimate
            Intercept    200.623
         Student[Yes]    476.676
               Income      6.218
  Student[Yes]:Income     -1.999
```

Contrast of marginal effects



Marginal Effects Estimation

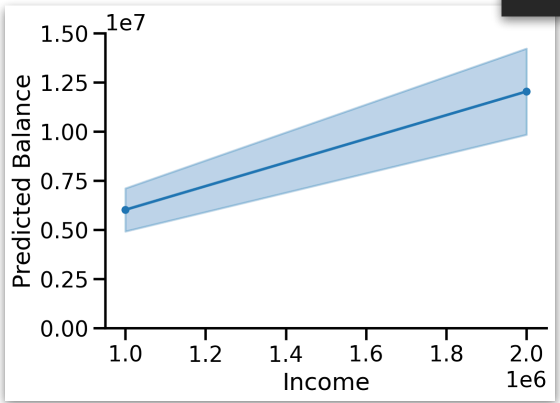
- Model-based predictions or comparisons between predictions evaluated over a range of predictor values (slider/switch settings)
  - *how does predicted-balance change if X switches from student to non-student?*
- Unifying framework for post-estimation analysis (e.g. post-hoc t-tests) via model interpolation
  - Not just comparisons for data you have, you can ask **what-if** questions!
  - *what if Income was $1mil? $2mil?*

Marginal Effects Recipe

what if Income was $1mil? $2mil?

- **What are we computing?**
  - predict balance
- **What values of predictors do we use?**
  - `Income@[1_000_000, 2_000_000]`
- **How do we aggregate?**
  - averaged over Student

```
m.explore("Income@[1000000, 2000000]")
```



Income	estimate
i64	f64
1000000	6.019e6
2000000	1.204e7

Model predicts that millionaires have millions in credit card balance!

Marginal Effects Recipe

what if Income was $1mil? $2mil? does this vary by student?

- **What are we computing?**
  - predict balance
- **What values of predictors do we use?**
  - `Income@[1_000_000, 2_000_000]`
- **How do we aggregate?**
  - averaged **within** Student

```
m.explore("Student ~ Income@[1000000, 2000000]")
```

Income	Student	estimate
f64	str	f64
1e6	"No"	6.218e6
1e6	"Yes"	4.22e6
2e6	"No"	1.244e7
2e6	"Yes"	8.439e6

Marginal Effects Recipe

what if Income was $1mil? $2mil? is this different by student?

- **What are we computing?**
  - predict balance **difference**
- **What values of predictors do we use?**
  - `Income@[1_000_000, 2_000_000]`
- **How do we aggregate?**
  - averaged by Income Levels

```
m.explore("Student[Yes - No] ~ Income@[1000000, 2000000]")
```

Income	contrast	estimate
f64	str	f64
1e6	"StudentYes - StudentNo"	-1.999e6
2e6	"StudentYes - StudentNo"	-3.998e6

Marginal Effects Estimation

- Model-based predictions or comparisons between predictions evaluated over a range of predictor values (slider/switch settings)
- **marginal means:** predictions for categorical-predictor levels when continuous-predictors are set to their means
- **marginal slopes:** predictions for continuous predictors averaged over categorical predictor levels OR differences between marginal means
- **Intuition:** we're just seeing how predictions $\hat{y}$ change when we wiggle around X using the $\hat{\beta}$ we estimated!

Marginal Effects Caveats

- Marginal effects **fully depend on your estimated model:** if your model is wrong then interpretation will be too!
- However, they **do not depend** upon categorical encoding you used!
  - This is **why** they're so useful for complex models (e.g. interactions, non-linearities)!
- Two notions of marginal effect for categorical terms to be aware of
  - **MEM**: marginal effect at the mean
  - **AME**: average marginal effect

MEM: Marginal Effect at the Mean

$$MEM = \frac{\delta \hat{y}}{\delta x} \text{ evaluated at } \bar{x}_1, \bar{x}_2, \ldots$$

- Compute the effect of X while holding all other terms at their **mean value**
  - Plug the mean of every other predictor into the model
  - Compute the **slope** at that point
- **Mixing-board Intuition**:
  - Like setting every slider to its average position, then *nudging* one slider and seeing what happens

AME: Average Marginal Effect

$$AME = \frac{1}{N}\sum_{i=1}^{N}\frac{\delta \hat{y}}{\delta x}$$

- Compute the effect of X **separately** for each observation, *then* **average** those effects
  - Let every data point keep their *own* covariates and compute their *own* **slope** at that point
  - Average those changes over observations
- **Mixing-board Intuition**:
  - Imagine we had separate versions ("tracks") of a single song. We adjust sliders for each track separately, then average tracks for the final version

MEM vs AME

$$MEM = \frac{\delta \hat{y}}{\delta x} \text{ evaluated at } \bar{x}_1, \bar{x}_2, \ldots \qquad AME = \frac{1}{N}\sum_{i=1}^{N}\frac{\delta \hat{y}}{\delta x}$$

- Will **always** be the same in **simple additive** models
  - *No interaction terms & No non-linearities*
  - **In math:** when the model's prediction(s) is **constant** across covariate space
- When we have **unbalanced data** (e.g. categorical predictor with uneven group-sizes) **how we average** will change if MEM = AME
  - Using observed group-sizes = **equal**
  - Weighting by group-size = **unequal**

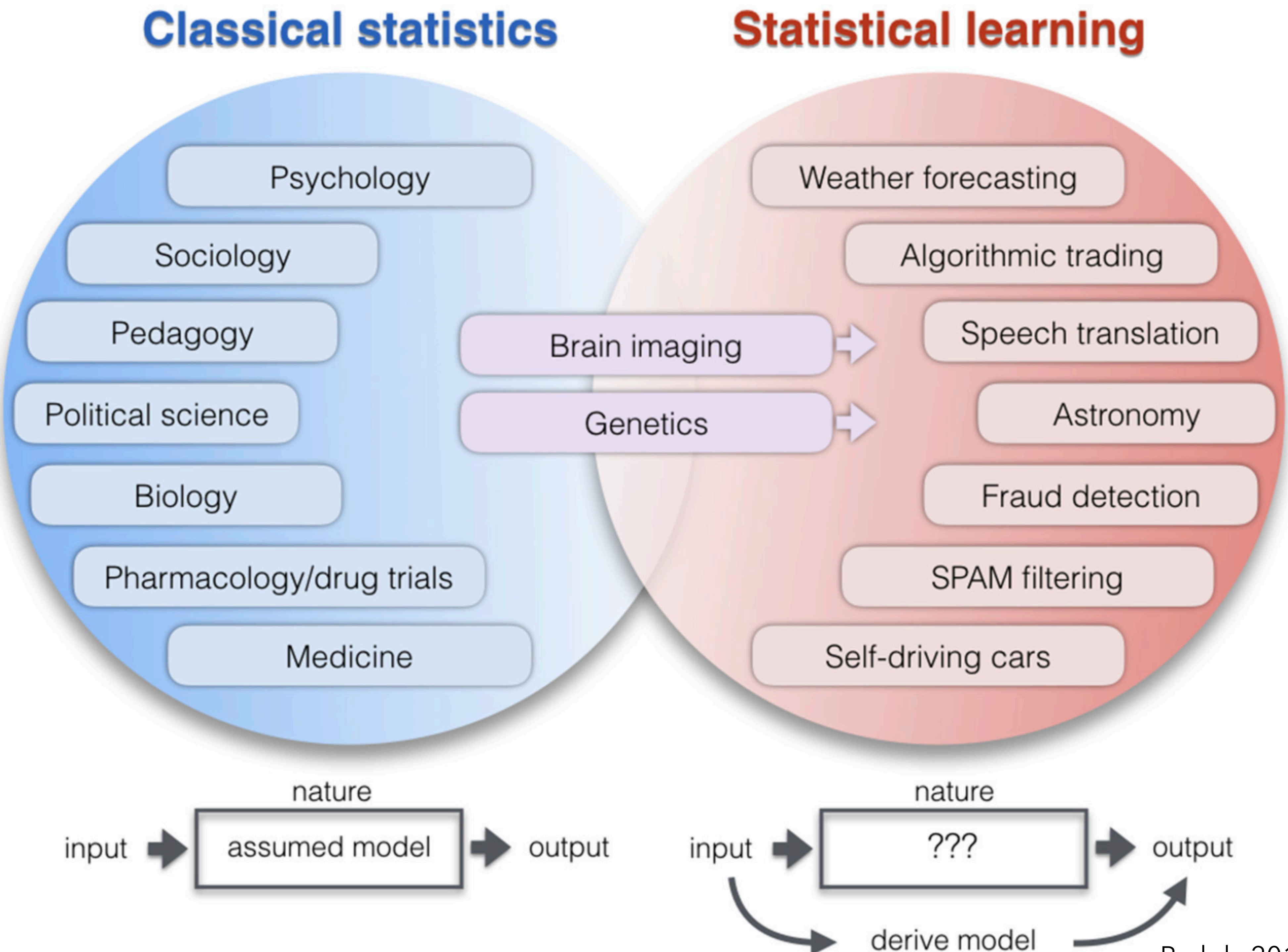
MEM vs AME

$$MEM = \frac{\delta \hat{y}}{\delta x} \text{ evaluated at } \bar{x}_1, \bar{x}_2, \ldots \qquad AME = \frac{1}{N}\sum_{i=1}^{N} \frac{\delta \hat{y}}{\delta x}$$

- Are not equal in **non-linear** and **complex** models
  - General-ized Linear Model (e.g. logistic regression)
  - Linear Mixed Models with unbalanced data
- We'll explore this in tomorrow's notebook!

Wrap Up

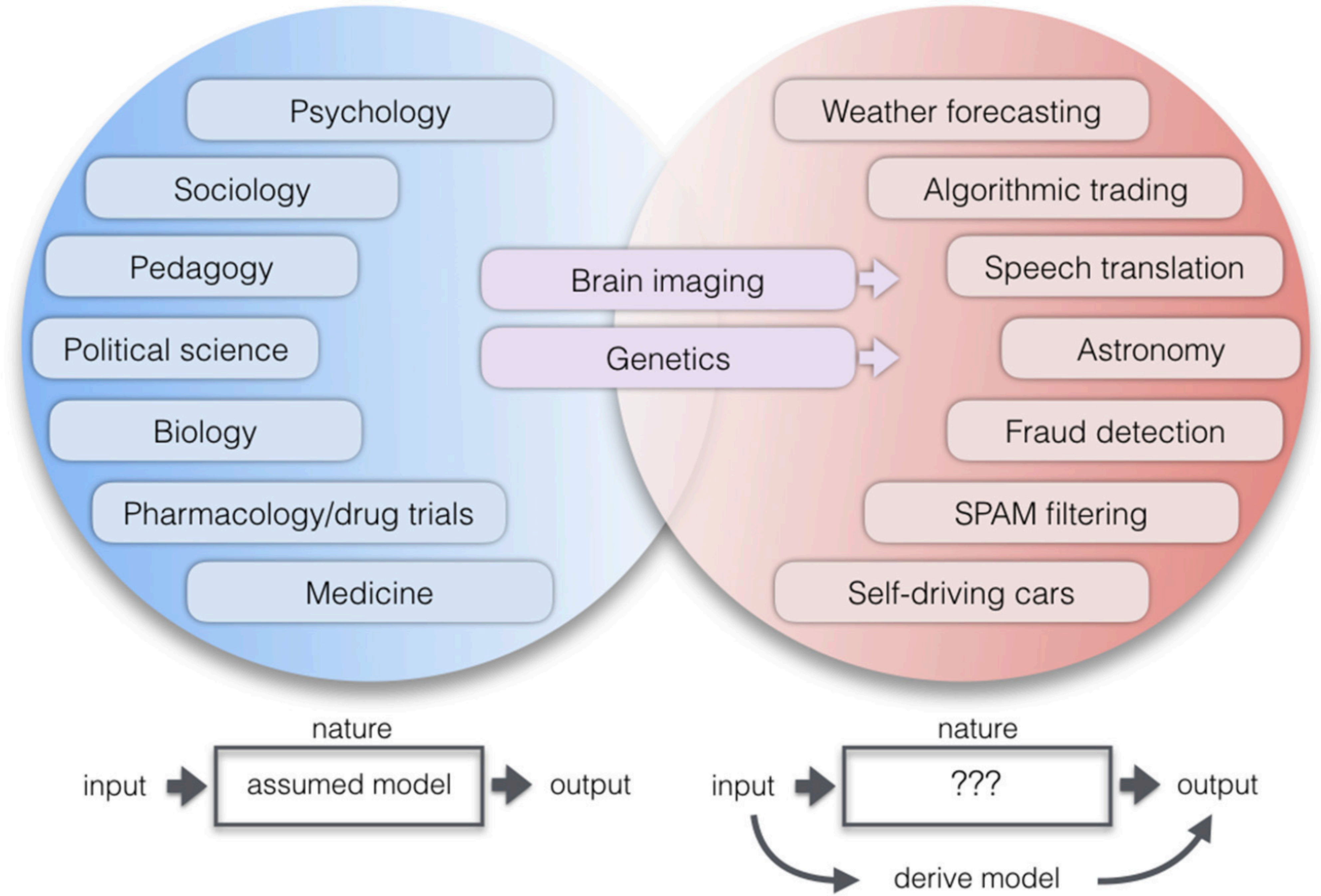
- Marginal effects allow you to use a model to **explore questions**
  - *How would predictions change if…*
- They allow you to **translate** a statistical model into **meaningful scientific statements**
- Your research question is rarely captured in raw parameter estimates!
- Think of parameter estimates as *intermediate artifacts* — now that you have them **use them**


Classical statistics
Statistical learning
Psychology
Sociology
Pedagogy
Political science
Biology
Pharmacology/drug trials
Medicine
Brain imaging
Genetics
Weather forecasting
Algorithmic trading
Speech translation
Astronomy
Fraud detection
SPAM filtering
Self-driving cars
nature
input
assumed model
output
nature
input
???
output
derive model

Bzdok, 2017

Explanation

Prediction



Marginal Effects Estimation

Explanation ⟷ **Prediction**

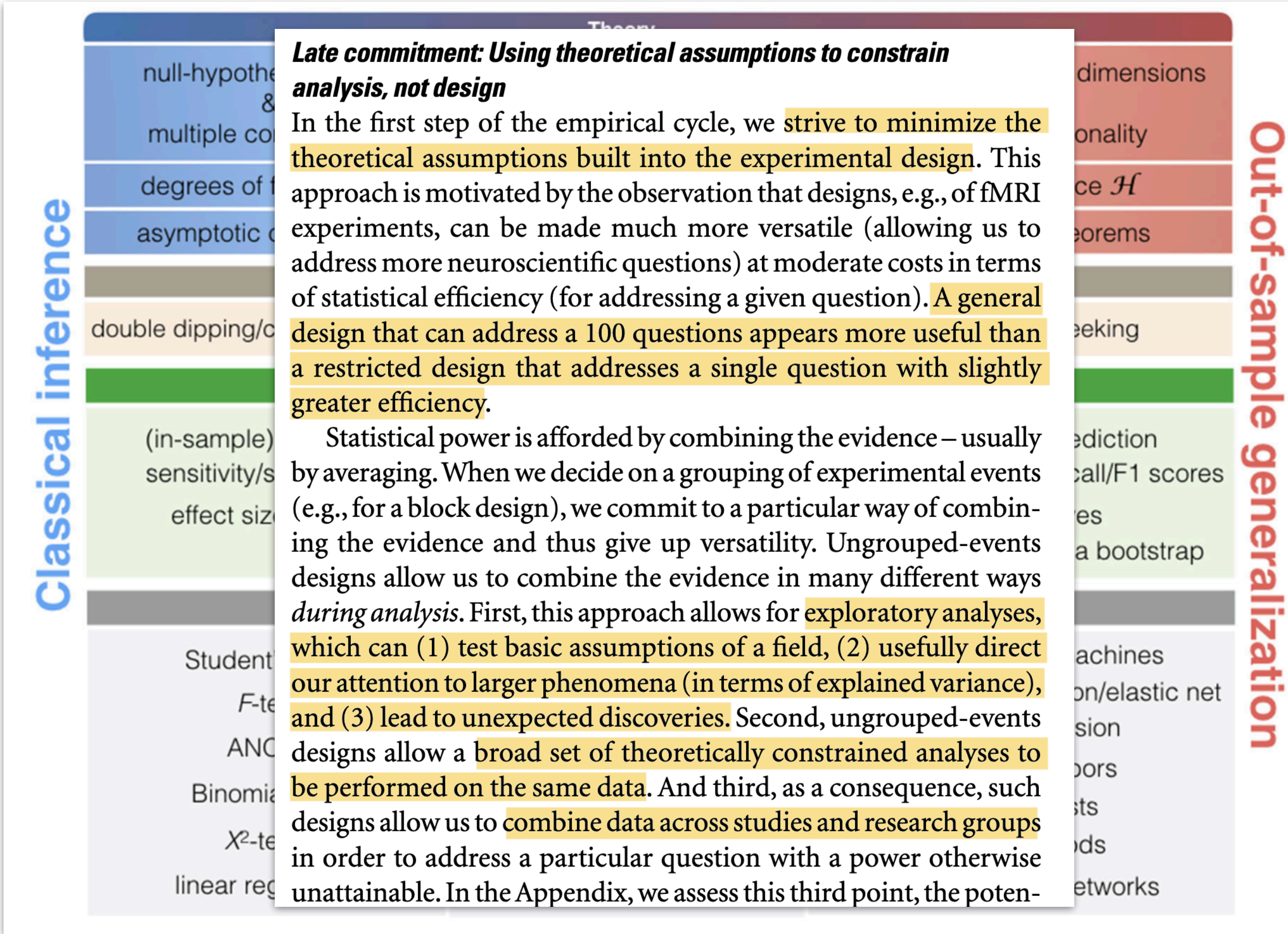
Classical inference | Out-of-sample generalization

Explanation	Marginal Effects Estimation	Prediction
Theory		
null-hypothesis testing & multiple comparisons	bias-variance decomposition	Vapnik-Chervonenkis dimensions & curse of dimensionality
degrees of freedom df		hypothesis space $\mathcal{H}$
asymptotic consistency		finite-sample theorems
Invalidations of inferential process		
double dipping/circular analysis	post-selection inference	data snooping/peeking
Outcome metrics		
(in-sample) p values sensitivity/specificity effect size/power	explained variance metrics AUC/ROC curve confidence intervals	out-of-sample prediction accuracy/precision/recall/F1 scores learning curves certainty estimates via bootstrap
Representative methods		
Student's *t*-test *F*-test ANOVA Binomial test X^2-test linear regression	general(ized) linear model	support vector machines LASSO/ridge regression/elastic net logistic regression nearest neighbors random forests kernel methods („deep") neural networks

Bzdok & Ionnadis, 2019

Marginal Effects Estimation

Explanation ⟷ **Prediction**



Late commitment: Using theoretical assumptions to constrain analysis, not design

In the first step of the empirical cycle, we strive to minimize the theoretical assumptions built into the experimental design. This approach is motivated by the observation that designs, e.g., of fMRI experiments, can be made much more versatile (allowing us to address more neuroscientific questions) at moderate costs in terms of statistical efficiency (for addressing a given question). A general design that can address a 100 questions appears more useful than a restricted design that addresses a single question with slightly greater efficiency.

Statistical power is afforded by combining the evidence – usually by averaging. When we decide on a grouping of experimental events (e.g., for a block design), we commit to a particular way of combining the evidence and thus give up versatility. Ungrouped-events designs allow us to combine the evidence in many different ways *during analysis*. First, this approach allows for exploratory analyses, which can (1) test basic assumptions of a field, (2) usefully direct our attention to larger phenomena (in terms of explained variance), and (3) lead to unexpected discoveries. Second, ungrouped-events designs allow a broad set of theoretically constrained analyses to be performed on the same data. And third, as a consequence, such designs allow us to combine data across studies and research groups in order to address a particular question with a power otherwise unattainable. In the Appendix, we assess this third point, the poten-

Bzdok & Ionnadis, 2019

Tomorrow

- Notebooks Review + new notebook
- **Please review readings on course website**
  - They'll help fill in details!
- **Rest of time:** Work on final project proposal